

Python Foundations Quiz

Test your fundamental knowledge of Python data types, control flow, functions, and standard concepts.

Name: _____

Date: _____

Time Allowed: 20 Minutes

Total Questions: 10

Q1. What is the output of the following Python expression?

```
print(type(5 / 2))
```

- ☐ A) <class 'int'>
- ☐ B) <class 'float'>
- ☐ C) <class 'number'>
- ☐ D) <class 'double'>

Q2. Which of the following data types in Python is mutable?

- ☐ A) Tuple
- ☐ B) String
- ☐ C) List
- ☐ D) Integer

Q3. What will be printed after executing this code block?

```
x = [1, 2, 3]
y = x
y.append(4)
print(len(x))
```

- ☐ A) 3
- ☐ B) 4
- ☐ C) AttributeError
- ☐ D) TypeError

Q4. What is the result of the expression `2 ** 3 ** 2` in Python?

- ☐ A) 64
- ☐ B) 512
- ☐ C) 36
- ☐ D) 12

Q5. How do you define a default parameter value in a Python function definition?

- ☐ A) `def func(val : 10):`
- ☐ B) `def func(val := 10):`
- ☐ C) `def func(val = 10):`
- ☐ D) `def func(default val = 10):`

Q6. What is the value of `result` after executing the list comprehension?

```
result = [x for x in range(5) if x % 2 == 0]
```

- ☐ A) [0, 2, 4]
- ☐ B) [1, 3]
- ☐ C) [0, 1, 2, 3, 4]
- ☐ D) [2, 4]

Q7. Which keyword is used to handle exceptions in Python?

- ☐ A) `catch`
- ☐ B) `except`
- ☐ C) `try_catch`
- ☐ D) `error`

Q8. What does the `dict.get(key, default)` method do if the key is NOT found in the dictionary?

- ☐ A) Raises a `KeyError`
- ☐ B) Returns `None` always
- ☐ C) Returns the provided `default` value
- ☐ D) Adds the key with default value to dict

Q9. What is the output of slicing `"Python"[1:4]` ?

- ☐ A) 'Pyt'
- ☐ B) 'yth'
- ☐ C) 'ytho'
- ☐ D) 'pyt'

Q10. Which statement best describes a Python `set` ?

- ☐ A) An ordered collection of key-value pairs
- ☐ B) An unordered collection of unique elements
- ☐ C) An ordered, immutable list of items
- ☐ D) A sequence indexed by integer positions

Answer Key & Explanations

Q1: B (<class 'float'>)

In Python 3, the single forward slash `/` represents true division and always returns a `float`, even if the result is a whole number. Integer division is performed using `//`.

Q2: C (List)

Lists are mutable object sequences, meaning their contents can be modified after creation (e.g., adding, removing, or changing elements). Tuples, Strings, and Integers are immutable.

Q3: B (4)

Assigning `y = x` creates a new reference pointing to the exact same list object in memory. Modifying `y` via `append(4)` also affects `x`, so `len(x)` becomes 4.

Q4: B (512)

Exponentiation operators `**` associate from right to left in Python. Thus, `2 ** 3 ** 2` is evaluated as `2 ** (3 ** 2)` = `2 ** 9` = 512.

Q5: C (def func(val = 10):)

Default parameter values are specified by assigning a value using the single equals operator `=` directly in the function definition header.

Q6: A ([0, 2, 4])

`range(5)` yields `0, 1, 2, 3, 4`. The conditional `if x % 2 == 0` filters for even integers, resulting in `[0, 2, 4]`.

Q7: B (except)

Python uses the `try ... except` block layout to catch and handle exceptions (unlike languages like Java/JavaScript which use `try ... catch`).

Q8: C (Returns the provided default value)

The `.get(key, default)` method safely looks up keys without raising a `KeyError`. If the key is absent, it returns the provided `default` argument (or `None` if no default is specified).

Q9: B ('yth')

String slicing `[start:stop]` includes index `start` and excludes index `stop`. In `"Python"`, index 1 is `'y'`, 2 is `'t'`, and 3 is `'h'` (index 4 `'o'` is excluded).

Q10: B (An unordered collection of unique elements)

A Python `set` is defined as an unordered collection of unique, hashable items. Sets automatically remove duplicates and do not support indexing.